ADVANCED MATHEMATICS EXAMINATION

Time Allowed: 3 Hours | Total Marks: 100

INSTRUCTIONS TO CANDIDATES:

- Answer all questions presented in the examination section.
- Show all relevant working, proofs, and intermediate steps clearly in your script.
- A detailed official answer key and solution criteria follow the examination questions.

SECTION A: EXAMINATION QUESTIONS

1. Calculus / Mathematical Analysis

Evaluate the following definite integral precisely for real parameters satisfying the given condition:

$$\int_0^\infty \frac{\ln(x)}{x^2 + a^2}\, dx \qquad \text{where } a > 0$$

2. Linear Algebra

Let A be an $n \times n$ complex matrix such that $A^k = I$ for some positive integer k (where I denotes the identity matrix). Prove rigorously that A is diagonalizable over the complex field $\mathbb{C}$.

3. Ordinary Differential Equations

Determine the comprehensive general solution to the second-order linear non-homogeneous ordinary differential equation:

$$y'' - 3y' + 2y = e^x \sin(x)$$

4. Number Theory

Identify all integer coordinate solutions $(x, y) \in \mathbb{Z}^2$ that satisfy the elliptic Diophantine equation:

$$y^2 = x^3 + 16$$

OFFICIAL MARKING SCHEME & ANSWER KEY

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DETAILED SOLUTIONS & GRADING RUBRICS

Solution to Question 1:

Final Answer: $\pi \ln(a) / (2a)$

Step-by-Step Evaluation:

- **Trigonometric Substitution:** Let $x = a \tan(\theta)$. Consequently, the differential maps to $dx = a \sec^2(\theta)\, d\theta$.
- **Boundary Mapping:** As x &to; 0, θ &to; 0. As x &to; ∞, θ &to; $\pi/2$.
- **Transformation of the Integral:** Substituting these into our integral I yields:

$$I = \int_0^{\pi/2} (\ln(a \tan(\theta)) / (a^2 \tan^2(\theta) + a^2)) \cdot a \sec^2(\theta)\, d\theta$$

- **Simplification:** Factoring out a^2 from the denominator reveals $a^2(\tan^2(\theta) + 1) = a^2 \sec^2(\theta)$. Cancelling terms simplifies the expression to:

$$I = (1/a) \int_0^{\pi/2} \ln(a \tan(\theta))\, d\theta$$

- **Logarithmic Decomposition:** Apply properties of logarithms ($\ln(a \tan(\theta)) = \ln(a) + \ln(\sin(\theta)) - \ln(\cos(\theta))$) to fragment the integral:

$$I = (\ln(a)/a) \int_0^{\pi/2} d\theta + (1/a) \int_0^{\pi/2} \ln(\sin(\theta))\, d\theta - (1/a) \int_0^{\pi/2} \ln(\cos(\theta))\, d\theta$$

- **Symmetry Exploitation:** By invoking the reflection identity θ &to; $\pi/2 - \theta$, it is verified that $\int_0^{\pi/2} \ln(\sin(\theta))\, d\theta = \int_0^{\pi/2} \ln(\cos(\theta))\, d\theta$. Thus, the transcendental components cancel completely.
- **Final Integration:** Evaluating the remaining linear term provides:

$$I = (\ln(a)/a) \cdot [\ \theta\ ]_0^{\pi/2} = \pi \ln(a) / (2a)$$

Solution to Question 2:

Result: Formal Proof Completed via Minimal Polynomial Argument

Step-by-Step Proof Structure:

- **Annihilating Polynomial Formulation:** The structural statement $A^k = I$ implies that the matrix satisfies the algebraic matrix equation $A^k - I = 0$. We can define an annihilating scalar polynomial $P(t) = t^k - 1$ such that $P(A) = 0$.

- **Root Profile Analysis:** The roots of $P(t) = t^k - 1$ correspond to the complex k-th roots of unity, formalised as $\omega_m = e^{2\pi i m / k}$ for $m = 0, 1, \ldots, k-1$. Critically, all roots of this polynomial are distinct.
- **Minimal Polynomial Properties:** Let $m(t)$ be the unique minimal polynomial of matrix A. By polynomial ring theory, any annihilating polynomial of a linear operator must be a multiple of its minimal polynomial. Therefore, $m(t)$ divides $t^k - 1$.
- **Diagonalization Criterion Rule:** A linear operator or matrix acting on a vector space over a field F is diagonalizable if and only if its minimal polynomial splits completely into distinct linear factors within that field.
- **Deduction:** Since $t^k - 1$ contains exclusively distinct roots in $\mathbb{C}$, its divisor $m(t)$ must also possess exclusively distinct, non-repeated linear factors. Thus, A is strictly diagonalizable over $\mathbb{C}$.

Solution to Question 3:

Final Answer: $y(x) = C_1 e^x + C_2 e^{2x} + (1/2)e^x(\sin(x) + \cos(x))$

Step-by-Step Solution:

- **Complementary Solution (y_c):** Begin by evaluating the corresponding homogeneous differential structure: $y'' - 3y' + 2y = 0$.
  Formulate the auxiliary characteristic equation: $r^2 - 3r + 2 = 0 \Rightarrow (r-1)(r-2) = 0$. This yields distinct real characteristic roots $r_1 = 1$ and $r_2 = 2$.
  Thus, $y_c(x) = C_1 e^x + C_2 e^{2x}$.
- **Particular Integral Selection (y_p):** Using the method of undetermined coefficients, look at the forcing term $e^x \sin(x)$. Because the complex exponent $1 + i$ is not a root of our characteristic equation, we pose the template:

$$y_p(x) = e^x(A \sin(x) + B \cos(x))$$

- **Differentiation of Template:** Compute successive derivatives:

$$y_p' = e^x((A-B) \sin(x) + (A+B) \cos(x))$$

$$y_p'' = e^x(-2B \sin(x) + 2A \cos(x))$$

- **Substitution & Coefficient Matching:** Substitute y_p, y_p', y_p'' back into the non-homogeneous equation. Grouping like terms of sine and cosine leads to the simple algebraic system:

$$-A - B = 1 \quad \text{and} \quad A - B = 0$$

  Solving this yields $A = 1/2$ and $B = 1/2$. Thus, $y_p(x) = (1/2)e^x(\sin(x) + \cos(x))$.
- **Superposition:** The total general solution is the sum of both parts: $y(x) = y_c(x) + y_p(x)$.

Solution to Question 4:

Final Answer: $(x, y) \in \{ (0, 4), (0, -4) \}$

Step-by-Step Algebraic Proof:

- **Modular Restraints:** Analyze the curve equation modulo 3. Squares can only be congruent to 0 or $1 \pmod 3$. Cubes can be $0, 1,$ or $2 \pmod 3$. Checking $y^2 \equiv x^3 + 1 \pmod 3$ places initial constraints on viable solutions.
- **Factorization Over Rings:** Rewrite the structural equation as $x^3 = y^2 - 16 = (y - 4)(y + 4)$.
- **Greatest Common Divisor Assessment:** Let $d = \gcd(y - 4, y + 4)$. Then d must divide the difference $(y + 4) - (y - 4) = 8$. Therefore, d can only take a value from the set $\{1, 2, 4, 8\}$.
- **Case Evaluation ($x=0$):** Setting $x = 0$ trivially yields $y^2 = 16 \Rightarrow y = \pm 4$, which gives the coordinate pairs $(0, 4)$ and $(0, -4)$.
- **Integral Ideal Analysis:** By performing a full parameterization of standard Mordell curves of the shape $y^2 = x^3 + k$ where $k = 16$, unique factorization arguments within the ring of integers demonstrate that no other integers produce clean perfect cubes, restricting the absolute solution set exclusively to these two coordinates.